PORTFOLIO

Arturo Rangel Alvarez

Mechanical Engineering
University of Detroit Mercy
Graduation: Spring 2027
LinkedIn
U.S. Citizen

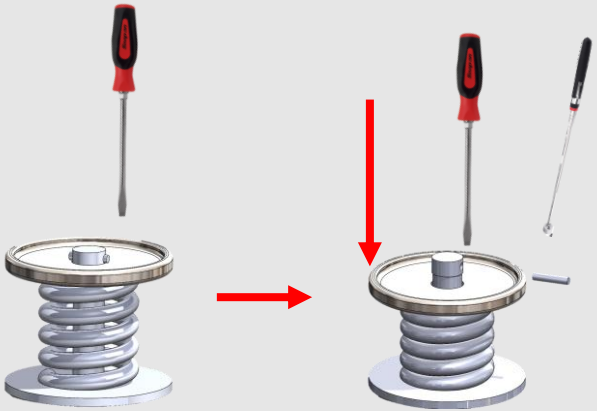
RENK AMERICA: TOOL DESIGN

Objective: Improve the efficiency and ergonomics of a manual disassembly process before scaling production from 5 to 40 transmissions/month.

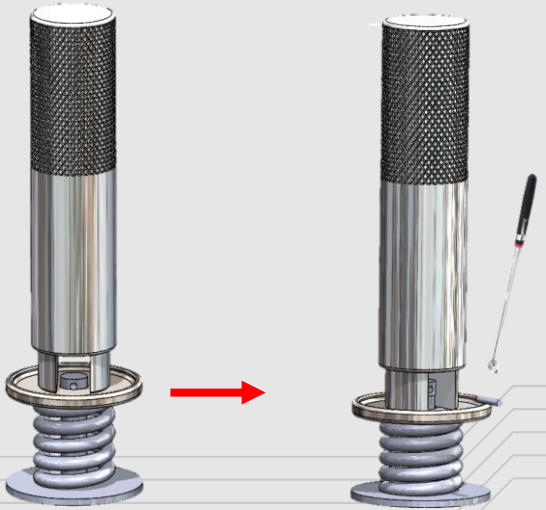
Approach: Designed an ergonomic tool in Creo, using mechanical leverage to reduce operator effort and improve contact with the assembly. Applied DFM principles, specified manufacturing tolerances, selected S2 tool steel, and performed FEA to evaluate the design.

Result: Reduced manual processing time by **21 minutes/day**, improved ergonomics, and reduced the risk of component damage during disassembly.

BEFORE



AFTER



(NON-PROPRIETARY DATA)

RENK America: CNC Machining Process Optimization

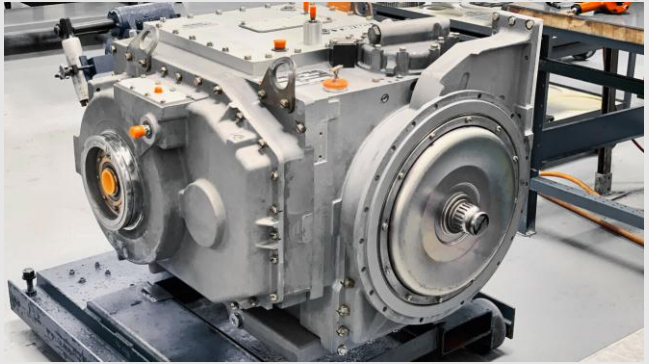
Objective: Identify and eliminate CNC machining bottlenecks to improve production efficiency and support higher manufacturing rates.

Approach: Analyzed machining parameters, tooling specifications, and G/M code for 100+ tools. Implemented three improvements:

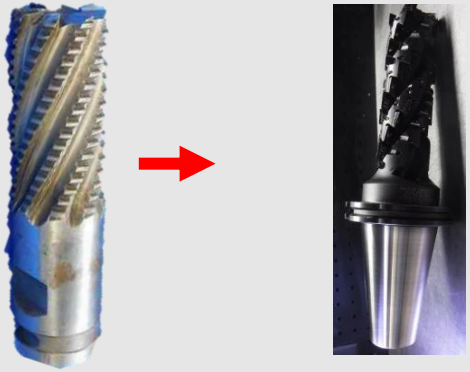
- **Milling:** Recreated toolpaths in Mastercam and replaced an HSS roughing mill with an indexable carbide helical rougher.
- **Deburring:** Selected a higher-performance CNC brush and validated increased feeds and speeds through surface-roughness inspections.
- **Tapping:** Optimized FANUC tapping-retract parameters to reduce non-cutting time.

Tested and validated process changes before production implementation.

Result: Saved 798 seconds of machining time per day across two transmission housings while maintaining dimensional quality.



HMPT Transmission Assembly



Previous HSS Cutting tool

Replacement Indexable Helical rougher

(REFERENCE-ONLY PICTURES)

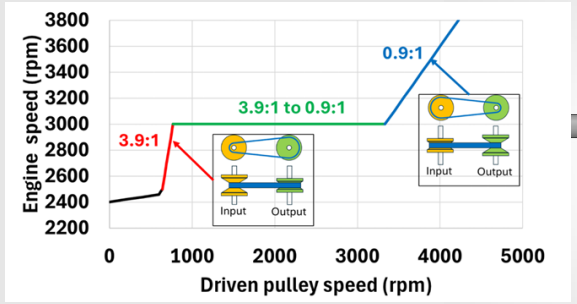
(NON-PROPRIETARY DATA)

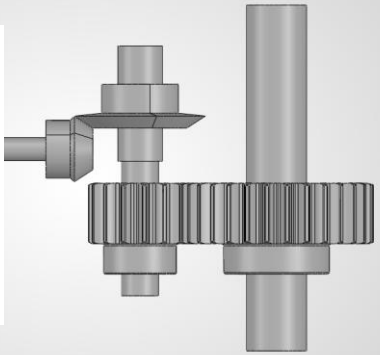
Baja SAE: Gearbox Design (In Progress)

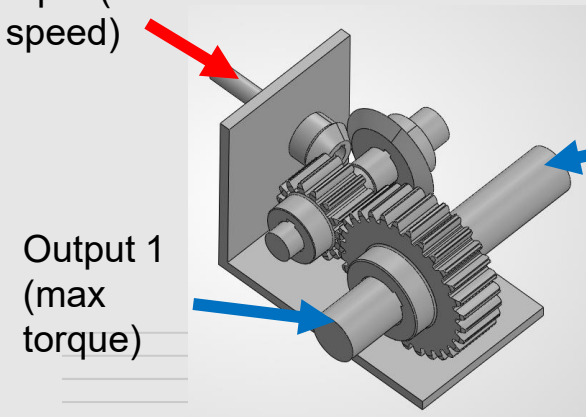
Objective: Lead the design and integration of a 4:1 gearbox for a Baja SAE 4WD vehicle, optimizing weight, packaging, and manufacturability.

Approach: Evaluated multiple gearbox architectures, selected commercial spiral bevel and spur gears, and performed gear strength, force, and safety factor analyses. Applying DFM/DFMA principles to CAD integration and housing development, considering shaft and bearing interfaces, sealing, machining constraints, and assembly requirements.

Result (In Progress): Developed two preliminary gearbox configurations, identifying an alternative that reduces gear mass by 8.3%. Quantified strength, cost, and packaging tradeoffs to guide final architecture selection and mechanical design.





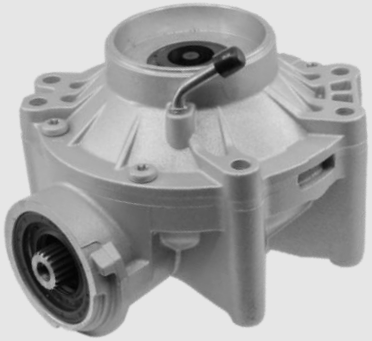


Baja SAE: Reverse Engineering & Vehicle Packaging

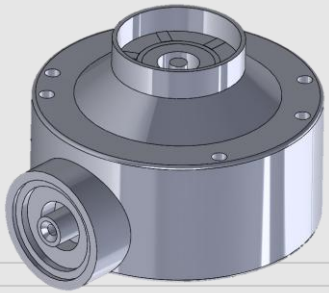
Objective: Create accurate CAD references of existing drivetrain and suspension components to support chassis design, vehicle packaging, and future component selection.

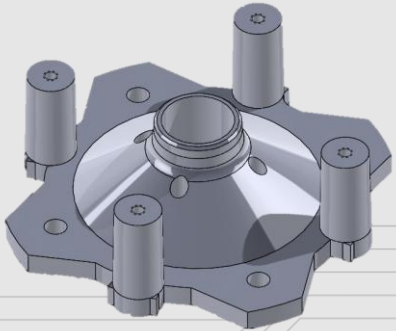
Approach: Reverse-engineered the rear differential and rear wheel hubs in SolidWorks, prioritizing critical mating holes, mounting interfaces, and overall dimensions rather than unnecessary geometric detail. Collaborated with teammates modeling tie rods, rack-and-pinion components, and front and rear knuckles to develop an integrated vehicle assembly.

Result: Developed accurate CAD references of drivetrain and suspension components, enabling vehicle-level integration, packaging evaluation, and verification of mounting interfaces before hardware procurement.







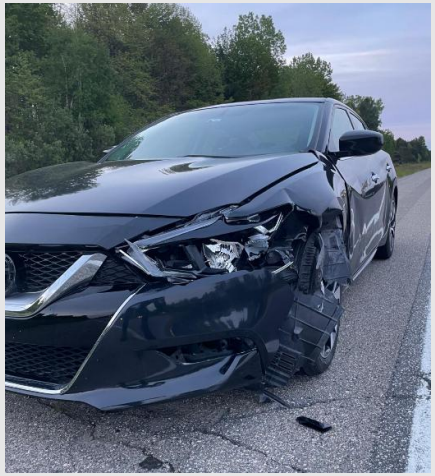


Automotive Restoration: Mechanical Assembly & Testing

Objective: Independently restore a collision-damaged 2016 Nissan Maxima through mechanical repair, component replacement, and functional verification.

Approach: Interpreted technical diagrams, disassembled damaged components, installed replacement hardware, and performed final assembly using hand tools. Pressure-tested the cooling system at 1.2× operating pressure and verified proper vehicle operation.

Result: Successfully restored the vehicle, verified cooling-system integrity without leaks, and confirmed normal operation without overheating.



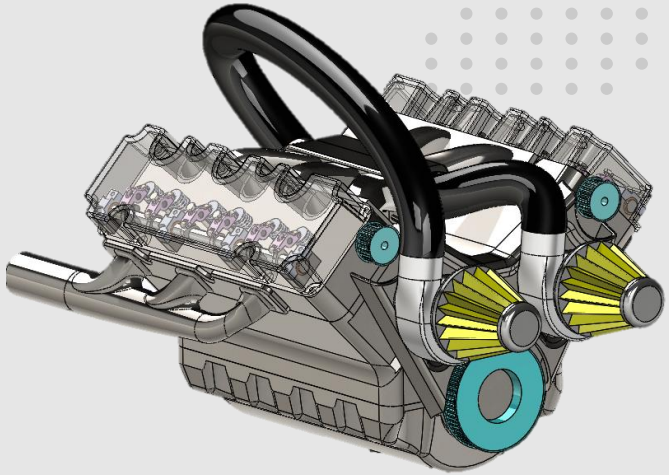


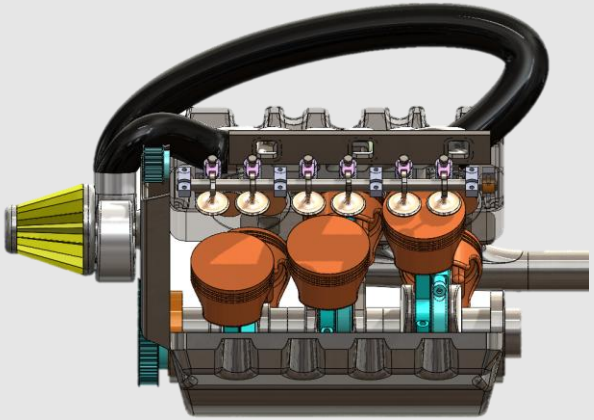




SOLIDWORKS MODEL AND ASSEMBLY: TWIN-TURBO V6

Modeled and assembled a twin-turbo V6 engine in SolidWorks, developing individual mechanical components and integrating them into a complete 3D assembly.







SOCIETY OF HISPANIC PROFESSIONAL ENGINEERS | VICE PRESIDENT

Community Involvement & Leadership

- Three-year SHPE member and chapter Vice President, connecting 20 students with industry professionals, networking opportunities, and career resources.
- Foster collaboration across university chapters, SHPE regions, and professional organizations to organize joint events and expand opportunities.
- Organized and delivered financial literacy workshops for students and parents, supporting Hispanic communities through education and professional development.







NCAA DIVISION I FENCING ATHLETE

11+ Years of High-Performance Competition

- Four seasons on the Mexican National Team, five-time national champion, and continental runner-up (2020).

Leadership Under Pressure

- Captain of a 25-athlete NCAA Division I team, leading by example and setting high standards on and off the fencing strip.

Building a Winning Team

- Maintain team morale, accountability, and focus under pressure, helping teammates overcome challenges and perform at their best.



